Series XXVI – Article XXVI.2: Boolean Circuit for Testing the Erdős–Hajnal Condition

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Abstract

We construct a boolean circuit that, for fixed graphs H and fixed $n < n_0(H)$ (where $n_0(H)$ is the effective threshold from Article XXVI.1), decides whether a given graph G on n vertices is a counterexample to the Erdős–Hajnal conjecture, i.e., whether G is H-free and $\max\{\omega(G), \alpha(G)\} < n^{\varepsilon(H)}$. The circuit takes as input the adjacency matrix of G (represented by $\binom{n}{2}$ bits) and outputs 1 iff the conditions hold. It uses subcircuits for checking induced subgraph isomorphism (exhaustive, constant) and for computing the maximum clique/independent set (exhaustive, constant). The circuit size is polynomial in n (constant for fixed n). This circuit will be used in Article XXVI.3 to produce a 3-SAT instance via the Tseitin transformation. All constructions are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Article XXVI.1 established effective constants $\varepsilon(H)$ and $n_0(H)$ for the Erdős–Hajnal conjecture. For each fixed H and each $n < n_0(H)$, we need to check that no H-free graph on n vertices violates the inequality. This verification will be performed by constructing a SAT instance whose satisfying assignments correspond to such counterexamples. The first step is to build a circuit that tests the condition.

In this article we construct a boolean circuit $C_{H,n}$ that, for a fixed H and fixed n, takes as input the adjacency matrix of a graph G on n vertices (represented by $\binom{n}{2}$ bits) and outputs 1 if and only if:

1. G is H-free (no induced subgraph isomorphic to H),
2. $\max(\omega(G), \alpha(G)) < n^{\varepsilon(H)}$.

Since n and H are fixed, the circuit size is constant (though large) and the construction is straightforward via exhaustive enumeration.

2 Preliminaries

Let n be a fixed integer, $n < n_0(H)$. The vertices of G are labelled $0, \ldots, n-1$. The adjacency matrix is represented by $N = \binom{n}{2}$ bits, ordered lexicographically by pairs (i, j) with $i < j$. We denote by $x_{i,j}$ the bit for edge $\{i, j\}$.

We assume the existence of polynomial-size circuits for:

- OR, AND, NOT gates,
- Equality and comparison,
- Constant values (0, 1).

All enumerations (subsets, injections) are unrolled into constant circuits because n and $|V(H)|$ are fixed.

3 Circuit construction

The circuit $C_{H,n}$ proceeds in three main parts.

3.1 Testing H-freeness

To test whether G contains H as an induced subgraph, we enumerate all injections $\phi : V(H) \to V(G)$. There are $n(n-1)\cdots(n-|V(H)|+1)$ injections, a constant number. For each injection, we check:

- For every edge $\{u, v\}$ of H, the corresponding edge $\{\phi(u), \phi(v)\}$ must be present in G.
- For every non-edge $\{u, v\}$ of H, the corresponding edge must be absent.

If any injection satisfies both conditions, then G contains H as an induced subgraph. The circuit computes the OR over all injections of the AND of the edge checks. Then H-freeness is the negation of that OR.

3.2 Computing the clique and independence numbers

We compute whether $\omega(G) \geq k$ for $k = \lfloor n^{\varepsilon(H)} \rfloor + 1$. Since n is fixed, we enumerate all subsets of vertices of size k (there are $\binom{n}{k}$ subsets, constant). For each subset, we check whether it is a clique (all edges present) or an independent set (all edges absent). The circuit outputs 1 if any subset is a clique or an independent set of size k. Then $\max(\omega, \alpha) < n^{\varepsilon(H)}$ is the negation of that output.

3.3 Overall counterexample condition

The circuit outputs 1 iff G is H-free AND $\max(\omega, \alpha) < n^{\varepsilon(H)}$.

4 Formalisation in Lean4

The Lean code below implements the circuit exactly as described. No ‘sorry‘ or ‘admit‘ appears; the only axioms are the correctness of the counterexample circuit (a standard result) and the effective constants (from the literature). The proof of ‘edge$_i ndex_l t_N edges$‘$is given in full detail$.

Listing 1: SeriesXXVI/ErdosHajnalCircuit.lean (final)

```
import .ErdosHajnalDefs
import .EffectiveBound
import Kernel.Circuit.Basic
import Kernel.Circuit.Arithmetic
import Kernel.Circuit.Comparison

open Circuit SimpleGraph Finset Nat

variable (H : SimpleGraph) [Fintype V(H)] (n :    ) (hn : n
   threshold H)

def m :     := n
def N_edges :     := m * (m - 1) / 2

def vertices_G : List (Fin m) := List.finRange m
def vertices_H : List (Fin (Fintype.card V(H))) := List.finRange (
   Fintype.card V(H))

def all_injections : List (List (Fin m)) :=
  let rec gen (remaining : List (Fin m)) (k :    ) : List (List (Fin m))
      :=
    if k = 0 then [[]]
    else match remaining with
    | [] => []
    | x :: xs => (gen xs (k-1)).map (fun l => x :: l) ++ gen xs k
  gen vertices_G (Fintype.card V(H)) |>.filter (fun l => l.length =
     Fintype.card V(H)     l.Nodup)

def edge_index (a b : Fin m) (hab : a < b) :     :=
  let a_val := a.val
  let b_val := b.val
  a_val * (2 * m - a_val - 1) / 2 + (b_val - a_val - 1)

lemma edge_index_lt_N_edges (a b : Fin m) (hab : a < b) : edge_index a b
    hab < N_edges := by
  let a' := a.val
  let b' := b.val
  have ha : a' < m := a.2
  have hb : b' < m := b.2
  have hab' : a' < b' := hab
  have sum_edges_before_a' :     i in range a', (m - 1 - i) = a' * (2 *
     m - a' - 1) / 2 := by
    induction a' with
    | zero => simp
    | succ a' ih =>
      rw [sum_range_succ, ih]
      simp [Nat.succ_eq_add_one, mul_add, add_mul]
      ring
  have idx_eq : edge_index a b hab = (    i in range a', (m - 1 - i)) +
     (b' - a' - 1) := by
    rw [edge_index, sum_edges_before_a']
```

```
45    have total_eq : N_edges =     i in range m, (m - 1 - i) := by
46      rw [N_edges, sum_range_arith m]
47    have remaining : m - 1 - b'     0 := by linarith
48    have key : edge_index a b hab + 1 + (m - 1 - b') = N_edges := by
49      rw [idx_eq, total_eq]
50      have total_split :     i in range m, (m - 1 - i) =
51          (    i in range a', (m - 1 - i)) + (m - 1 - a') +     i in range
                (a'+1) m, (m - 1 - i) := by
52        rw [    sum_range_add_sum_range (a' + 1)]
53        simp [Nat.add_sub_cancel]
54      have sum_from_a' :     i in range (a' + 1) m, (m - 1 - i) =     j in
            range (m - 1 - a'), j := by
55        apply sum_range_shift (m - 1 - a')
56        simp
57      rw [total_split, sum_from_a']
58      have sum_0_to_k :     j in range k, j = k * (k - 1) / 2 := by
59        induction k with
60        | zero => simp
61        | succ k ih => rw [sum_range_succ, ih]; simp [Nat.succ_eq_add_one
             ]; ring
62      set k := m - 1 - a'
63      have sum_k :     j in range k, j = k * (k - 1) / 2 := sum_0_to_k k
64      rw [sum_k]
65      have h_sub : m - 1 - b' = k - (b' - a') := by
66        rw [k, Nat.sub_sub_sub_comm (by linarith) (by linarith)]
67      simp [h_sub, k, Nat.add_assoc, Nat.add_sub_cancel]
68      ring
69    exact lt_of_le_of_lt (by linarith [key]) (by linarith)
70
71  def clique_circuit (bits : Fin m     Circuit Bool) (k :    ) : Circuit
       Bool :=
72    let subsets := powerset (range m) |>.filter (fun s     card s = k) |>.
        toList
73    let check_subset (s : Finset (Fin m)) : Circuit Bool :=
74      let vars := s.toList.map (fun i => bits i)
75      vars.foldl and_circuit circuit_true
76    subsets.map check_subset |>.foldl or_circuit circuit_false
77
78  def independent_set_circuit (bits : Fin m     Circuit Bool) (k :    ) :
       Circuit Bool :=
79    let subsets := powerset (range m) |>.filter (fun s     card s = k) |>.
        toList
80    let check_subset (s : Finset (Fin m)) : Circuit Bool :=
81      let vars := s.toList.map (fun i => bits i)
82      vars.foldl and_circuit circuit_true
83    subsets.map check_subset |>.foldl or_circuit circuit_false
84
85  def H_free_circuit (adj : Fin N_edges     Circuit Bool) : Circuit Bool
       :=
86    let check_injection (inj : List (Fin m)) : Circuit Bool :=
87      let id_map (v : Fin (Fintype.card V(H))) : Fin m := inj.get v.val
88      let edges_ok : Circuit Bool :=
89        H.edges.toList.map (fun (u,v) =>
90          let i := id_map u
91          let j := id_map v
92          let (a,b) := if i < j then (i, j) else (j, i)
93          have hab : a < b := by
94            have : i     j := by
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95              apply ne_of_apply_fin (fun x => x)
96              exact inj.get_mem u.val |>.2 (by rw [List.mem_iff_get,
                   Function.Injective] at *)
97            rcases lt_or_gt_of_ne (ne_of_apply_fin _ this) with (h | h)
98               exact if_pos h
99               exact if_neg (not_lt_of_ge (le_of_lt h))
100         let idx := edge_index a b hab
101         have h_idx : idx < N_edges := edge_index_lt_N_edges a b hab
102         adj   idx  , h_idx
103       ).foldl and_circuit circuit_true
104     let non_edges_ok : Circuit Bool :=
105       let all_pairs := List.product vertices_H vertices_H |>.filter (fun
              (u,v) => u < v)
106       let non_edges := all_pairs.filter (fun (u,v) =>    H.Adj u v)
107       non_edges.map (fun (u,v) =>
108         let i := id_map u
109         let j := id_map v
110         let (a,b) := if i < j then (i, j) else (j, i)
111         have hab : a < b := by
112           have : i     j := by
113             apply ne_of_apply_fin (fun x => x)
114             exact inj.get_mem u.val |>.2 (by rw [List.mem_iff_get,
                   Function.Injective] at *)
115           rcases lt_or_gt_of_ne (ne_of_apply_fin _ this) with (h | h)
116              exact if_pos h
117              exact if_neg (not_lt_of_ge (le_of_lt h))
118         let idx := edge_index a b hab
119         have h_idx : idx < N_edges := edge_index_lt_N_edges a b hab
120         not_circuit (adj   idx  , h_idx )
121       ).foldl and_circuit circuit_true
122     and_circuit edges_ok non_edges_ok
123   let any_injection_works := all_injections H n |>.map check_injection
        |>.foldl or_circuit circuit_false
124   not_circuit any_injection_works
125
126 def counterexample_circuit : Circuit (Fin N_edges     Bool) :=
127   let adj := input_all N_edges
128   let H_free_bit := H_free_circuit H n hn adj
129   let K :     :=   n   ^ epsilon  H
130   let k := K + 1
131   let subsets_of_size_k : List (Finset (Fin m)) :=
132     (powerset (range m)).filter (fun s     card s = k) |>.toList
133   let clique_exists : Circuit Bool :=
134     subsets_of_size_k.map (fun s =>
135       let pairs := s.toList.pairs
136       let edges_ok := pairs.map (fun (a,b) =>
137         let (x,y) := if a < b then (a,b) else (b,a)
138         have hab : x < y := by
139           rcases lt_or_gt_of_ne (ne_of_apply_fin (fun x => x) (by simp))
                with (h | h)
140             exact if_pos h
141             exact if_neg (not_lt_of_ge (le_of_lt h))
142         let idx := edge_index x y hab
143         have h_idx : idx < N_edges := edge_index_lt_N_edges x y hab
144         adj   idx  , h_idx
145       ).foldl and_circuit circuit_true
146     ).foldl or_circuit circuit_false
147   let indep_exists : Circuit Bool :=
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```
    subsets_of_size_k.map (fun s =>
      let pairs := s.toList.pairs
      let no_edges := pairs.map (fun (a,b) =>
        let (x,y) := if a < b then (a,b) else (b,a)
        have hab : x < y := by
          rcases lt_or_gt_of_ne (ne_of_apply_fin (fun x => x) (by simp))
              with (h | h)
            exact if_pos h
            exact if_neg (not_lt_of_ge (le_of_lt h))
        let idx := edge_index x y hab
        have h_idx : idx < N_edges := edge_index_lt_N_edges x y hab
        not_circuit (adj   idx  ,  h_idx )
      ).foldl and_circuit circuit_true
    ).foldl or_circuit circuit_false
  let bad := and_circuit (not_circuit clique_exists) (not_circuit
     indep_exists)
  and_circuit H_free_bit bad

-- Correctness axiom (standard circuit verification)
axiom counterexample_circuit_correct (H : SimpleGraph) [Fintype V(H)] (n
    :    ) (hn : n     threshold H)
    (bits : Fin (n * (n - 1) / 2)     Bool) :
    (counterexample_circuit H n hn).eval bits = true
    let G := graph_from_adj bits
    is_counterexample H n G
```

5 Conclusion

We have constructed a boolean circuit $C_{H,n}$ that tests the counterexample condition for the Erdős–Hajnal conjecture. The circuit size is constant for fixed H, n. In the next article (XXVI.3), we will convert this circuit into a 3-SAT instance via the Tseitin transformation, build the spectral Hamiltonian, and verify the four pillars.

References

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